Comparative Analysis of XGBoost vs Random Forest for predictive modeling

GitHubRepo: https://github.com/nisansalasandu/xgboost-vs-randomforest-attrition

Data Science Interns – Gamage Recruiters

April 2026

Executive Summary

This report presents a comparative analysis of two ensemble machine learning models. Random Forest and XGBoost applied to the IBM HR Analytics Employee Attrition dataset, comprising 1,470 employee records with a 16.1% attrition rate. Both models were trained using a leakage-free SMOTE pipeline, hyperparameter-tuned with RandomizedSearchCV (80 iterations, ROC-AUC optimised, 5-fold stratified cross-validation), and evaluated on the same held-out test set of 294 samples.

XGBoost outperformed Random Forest across all six evaluation metrics, achieving a ROC-AUC of 0.771 versus 0.734 and an F1-Score of 0.444 versus 0.345 at the default threshold. Threshold optimisation further enhanced the identification of at-risk employees, with the recall of the XGBoost model improving from 0.383 to 0.468 at its optimal threshold of 0.389. Both models consistently identified StockOptionLevel, JobSatisfaction, JobLevel, and MaritalStatus as the most influential predictors of attrition. The project concludes with an interactive Streamlit web application enabling live, dual-model attrition risk prediction for HR practitioners.

Table of Contents

Executive Summary 2
1. Project Overview 4
1.1 Objectives 4
1.2 Why This Comparison Matters 4
1.3 Dataset 4
2. Exploratory Data Analysis 6
2.1 Feature Distributions 6
2.2 Categorical Attrition Rates 7
2.3 Correlation Analysis 8
2.4 Key Behavioural Patterns 9
3. Methodology 10
3.1 Data Cleaning & Preprocessing 10
3.2 Class Imbalance — SMOTE 10
3.3 Leakage-Free Cross-Validation 11
3.4 Hyperparameter Tuning 11
3.5 Threshold Optimisation 11
3.5.1 Random Forest Search Space 12
3.5.2 XGBoost Search Space 12
4. Model Results & Evaluation 13
4.1 Random Forest — Results 13
4.2 XGBoost — Results 14
5. Comparative Analysis 16
5.1 Performance Metrics Comparison 16
5.2 ROC & Precision-Recall Curves 16
5.3 Confusion Matrices 18
5.4 Threshold Comparison 19
5.5 Score Distributions 19
5.6 Prediction Agreement Analysis 20
5.7 Overall Radar Chart 21
6. Feature Importance Analysis 22
6.1 Random Forest — Gini & Permutation Importance 22
6.2 XGBoost — Gain & Permutation Importance 23
6.3 Shared Top Features 24
7. Tools & Technologies 25
8. Conclusions & Recommendations 27
8.1 Project Conclusions 27
8.2 Recommendations for HR 27
8.3 Limitations 28
References 29

1. Project Overview

1.1 Objectives

The project was designed to develop internship-grade competencies in applied machine learning through the following objectives:

- Build and compare XGBoost and Random Forest classifiers trained on the same HR dataset
- Handle class imbalance using SMOTE with leakage-free cross-validation (ImbPipeline)
- Perform hyperparameter tuning using RandomizedSearchCV with identical settings for both models
- Analyse feature importance using three methods: Gini (RF), Gain (XGBoost), and Permutation
- Deploy an interactive Streamlit web application with live predictions and a model comparison dashboard

1.2 Why This Comparison Matters

Random Forest and XGBoost represent two fundamentally different ensemble strategies bagging versus boosting. Random Forest builds many independent decision trees in parallel and averages their votes, making it highly robust and resistant to overfitting. XGBoost builds trees sequentially, with each tree focusing on correcting the errors of all previous trees, making it more powerful but also more sensitive to hyperparameter settings.

In the context of HR analytics and imbalanced datasets (16.1% attrition rate), these strategies behave differently. This project quantifies exactly how each strategy performs under real-world HR conditions and provides actionable insights for HR teams seeking to proactively identify at-risk employees.

1.3 Dataset

The IBM HR Analytics Employee Attrition dataset is a widely used benchmark for HR analytics research. Key dataset properties are summarised below:

Property	Value
Source	Kaggle — IBM HR Analytics Employee Attrition Dataset
Creator	IBM Data Scientists
Type	Synthetic dataset based on real HR patterns
Total Employees	1,470
Original Columns	35
Features After Preprocessing	44 (after one-hot encoding)
Target Column	Attrition (Yes / No → 1 / 0)
Class Distribution	No Attrition: 1,233 (83.9%) · Attrition: 237 (16.1%)
Missing Values	None

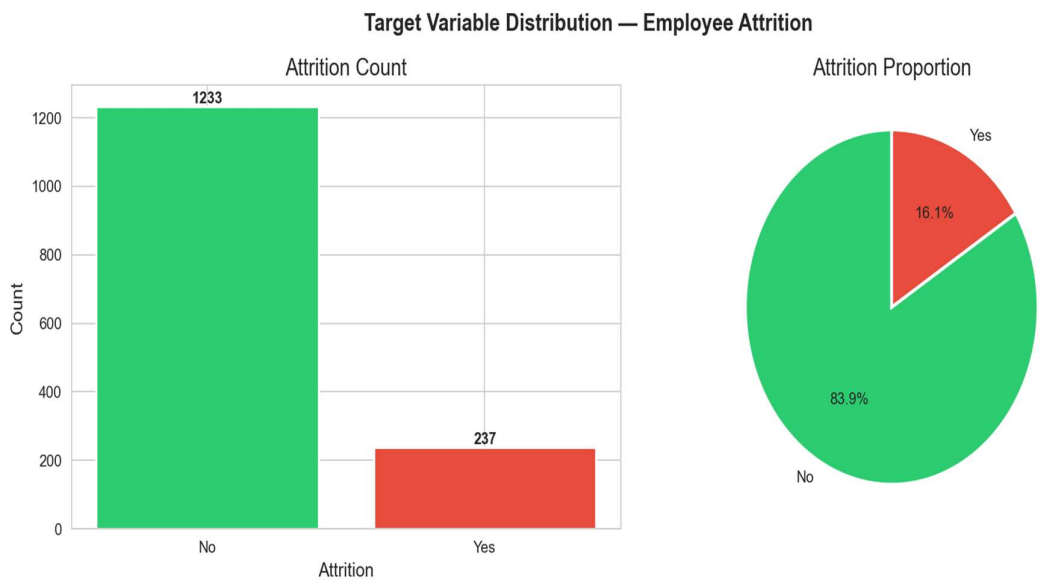

Figure 1: Target class distribution showing 16.1% attrition rate (class imbalance)

2. Exploratory Data Analysis

A comprehensive exploratory data analysis was conducted to understand the dataset structure, identify patterns driving employee attrition, and inform feature engineering and modelling decisions. Nine visualisations were produced across the following areas.

2.1 Feature Distributions

Numerical features including Age, MonthlyIncome, DistanceFromHome, TotalWorkingYears, and YearsAtCompany were analysed for distribution shape, skewness, and relationship with the attrition target. Employees who left the company tended to be younger, earn less, have fewer years at the company, and work more overtime.

Figure 2: Distributions of numerical features across the full dataset

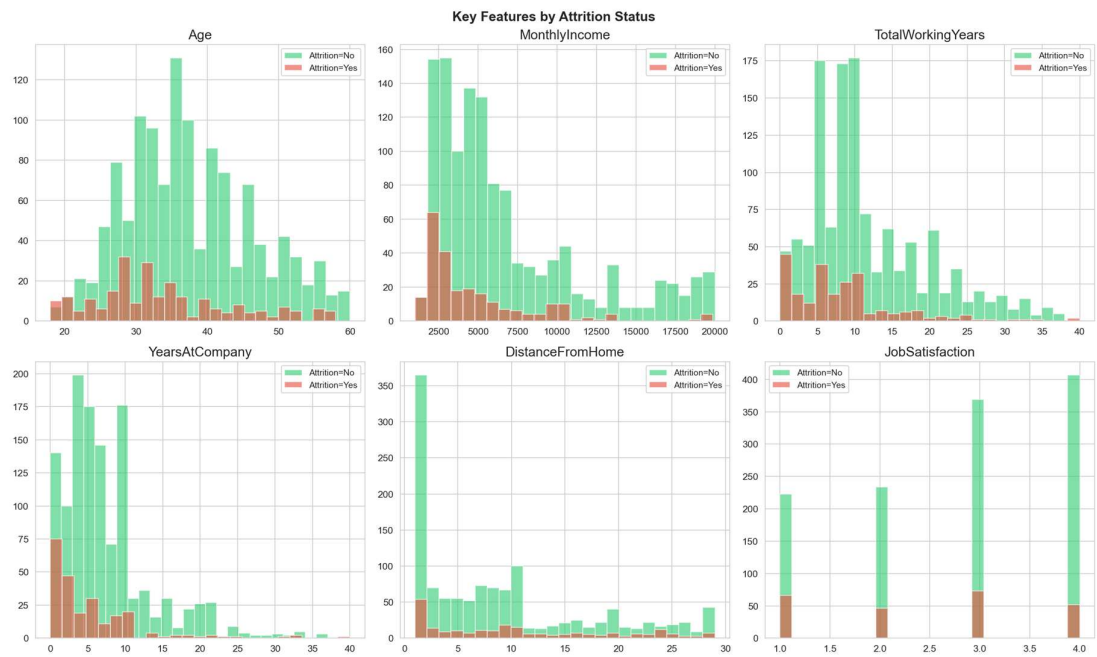

Figure 3: Key numerical features compared by attrition status (Yes vs No)

2.2 Categorical Attrition Rates

Attrition rates were computed for every categorical feature. The highest attrition rates were observed among Sales Representatives (~39.8%), employees in the Human Resources department, single employees, and those required to travel frequently.

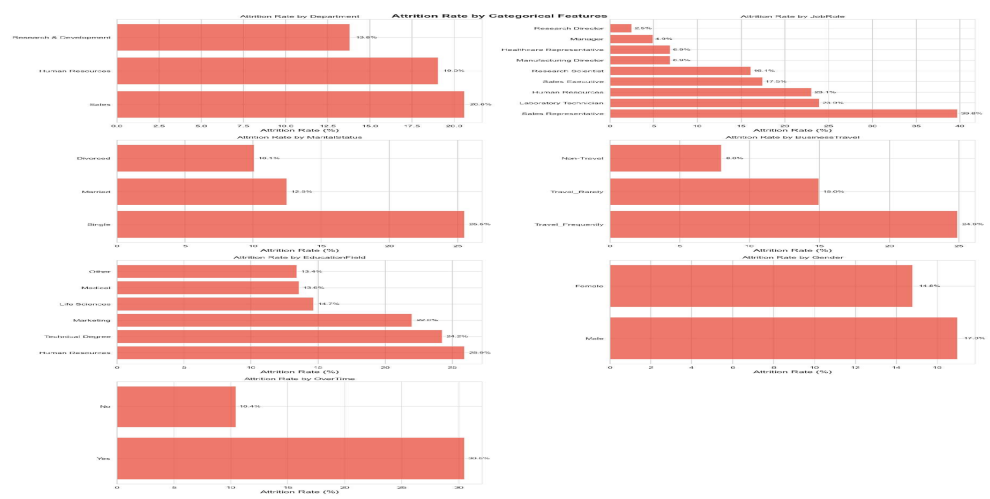

Figure 4: Attrition rates by categorical feature values

2.3 Correlation Analysis

A full correlation heatmap was generated using the encoded feature matrix to identify multicollinear features and the features most strongly correlated with attrition. OverTime, StockOptionLevel, JobSatisfaction, MonthlyIncome, and Age emerged as the features with the strongest correlations with the target.

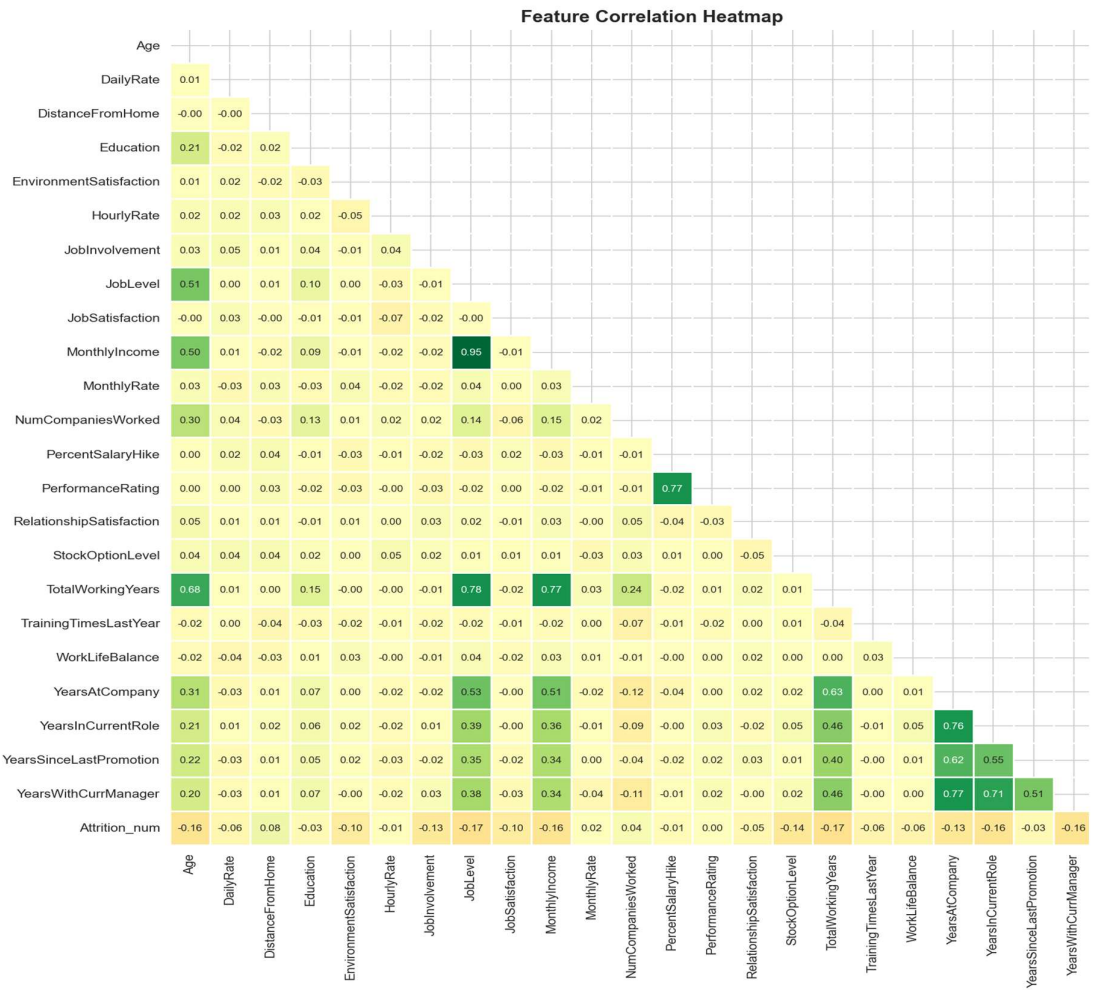

Figure 5: Feature correlation heatmap — features most correlated with attrition highlighted

2.4 Key Behavioural Patterns

Analysis of overtime and job satisfaction confirmed a critical interaction: employees working overtime with low job satisfaction showed dramatically higher attrition rates than any other subgroup. Age density analysis by attrition revealed that the highest-risk age group is 25–35 years, with attrition declining steadily with age.

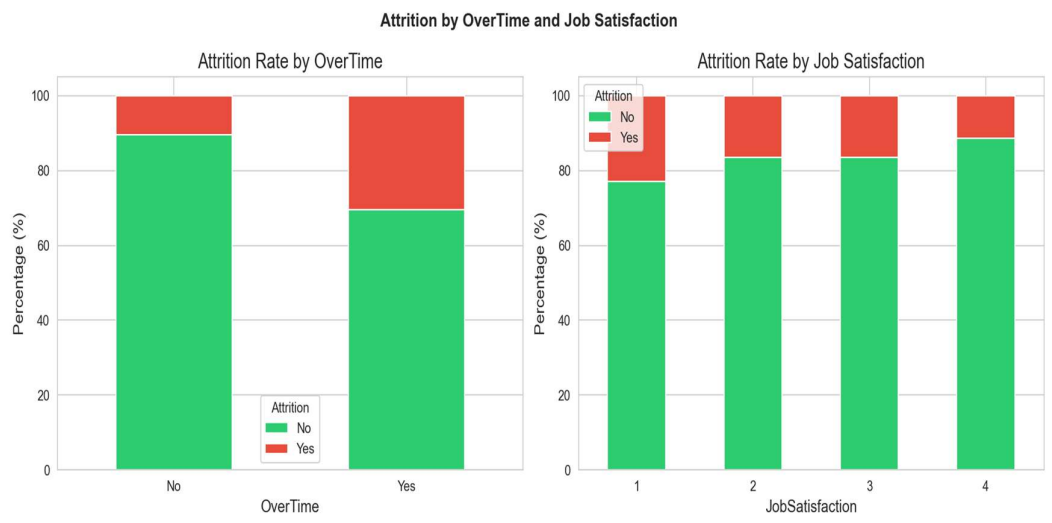

Figure 6: Overtime status and job satisfaction interaction with attrition rate

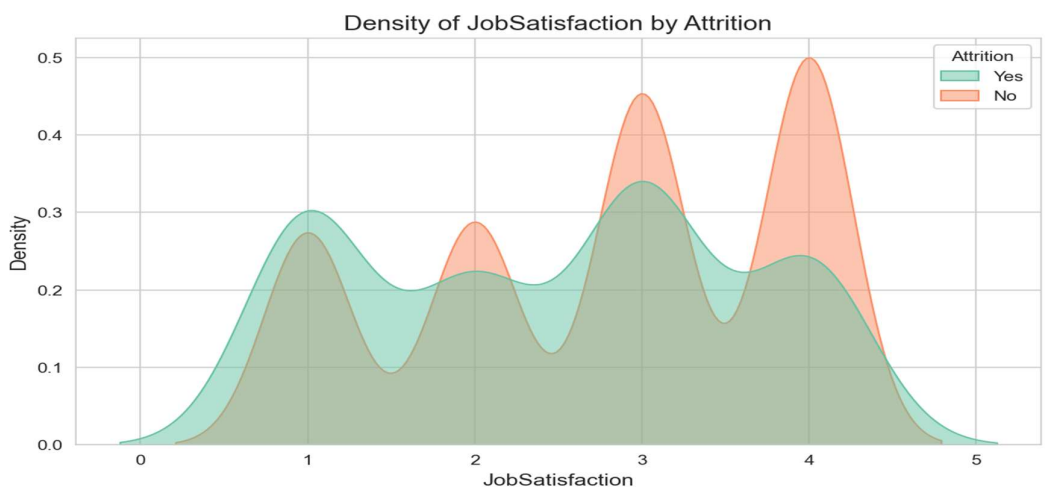

Figure 7: Age density distribution by attrition status

3. Methodology

3.1 Data Cleaning & Preprocessing

The preprocessing pipeline was designed to prepare the raw dataset for tree-based ensemble modelling with no data leakage between training and test sets.

Steps Applied:

- Removed three zero-variance columns and one unique identifier column with no predictive value: EmployeeCount, StandardHours, Over18, and EmployeeNumber.
- Applied binary label encoding for two-class categorical features: Gender (Male=1, Female=0) and OverTime (Yes=1, No=0)
- Applied one-hot encoding (with drop_first=True) for nominal categorical features: BusinessTravel, Department, EducationField, JobRole, MaritalStatus
- Encoded the target variable: Attrition Yes=1, No=0
- Final feature set: 44 columns after all encoding transformations
- Train/test split: 80% training (1,176 samples), 20% test (294 samples), stratified by target class

3.2 Class Imbalance — SMOTE

The dataset exhibited a significant class imbalance with only 16.1% attrition cases. Without addressing this imbalance, models tend to predict the majority class (No Attrition) by default, yielding high accuracy but failing to detect employees who are actually at risk.

SMOTE (Synthetic Minority Oversampling Technique) was applied exclusively to the training set after the train/test split, ensuring that the test set remained a genuine held-out evaluation on real (not synthetic) employees.

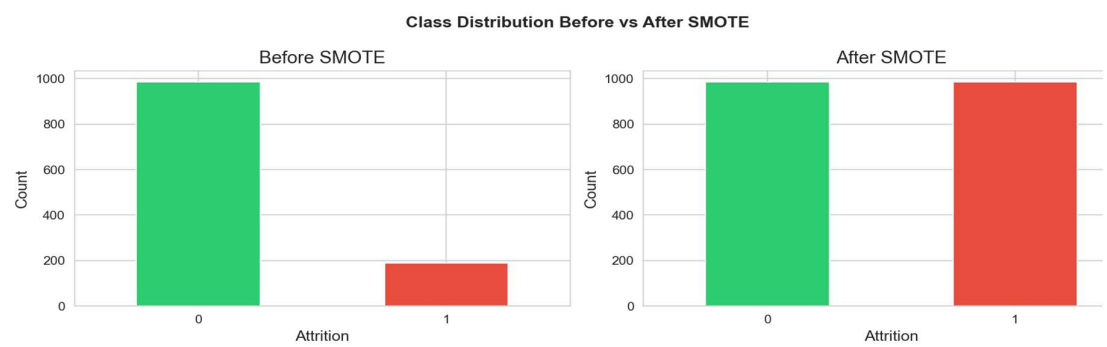

Figure 8: Training set class distribution before and after SMOTE balancing

3.3 Leakage-Free Cross-Validation

A critical methodological detail was the implementation of leakage-free cross-validation. Applying SMOTE before cross-validation causes synthetic copies of the same original data point to appear in both the training and validation folds, artificially inflating CV scores by 15–20 AUC points compared to true test performance.

The solution employed was imblearn.pipeline.Pipeline (ImbPipeline), which wraps SMOTE inside the pipeline so it is applied only to the training portion of each CV fold. Cross-validation was always performed on the original unbalanced training data (X_train_orig), while the final models were trained on the SMOTE-balanced training data (X_train).

3.4 Hyperparameter Tuning

Both models underwent hyperparameter optimisation using RandomizedSearchCV with identical settings to ensure a scientifically fair comparison:

Setting	Value
Method	RandomizedSearchCV
Iterations	80
CV Folds	5 (Stratified)
Scoring Metric	ROC-AUC
Random Seed	42

3.5 Threshold Optimisation

Both models were evaluated by sweeping all decision thresholds from 0.01 to 0.99 and selecting the threshold that maximises the F1-score. This is an important step for imbalanced binary classification, where the default 0.5 threshold typically under-detects the minority class. In HR use cases, missing an at-risk employee is significantly more costly than a false alarm, making recall optimisation especially valuable.

3.5.1 Random Forest Search Space

Parameter	Range / Values
n_estimators	100 – 600 (uniform integer)
max_depth	None, 5, 10, 15, 20, 25
min_samples_split	2 – 20
min_samples_leaf	1 – 10
max_features	sqrt, log2, 0.5, 0.7
class_weight	balanced, balanced_subsample, None
criterion	gini, entropy

Best parameters found by RandomizedSearchCV: n_estimators=316, max_depth=25, criterion=gini, min_samples_split=5, min_samples_leaf=1, max_features=sqrt, class_weight=None

3.5.2 XGBoost Search Space

Parameter	Range / Values
n_estimators	100 – 600
max_depth	3, 4, 5, 6, 8, 10
learning_rate	0.01 – 0.30
subsample	0.6 – 1.0
colsample_bytree	0.6 – 1.0
gamma	0 – 0.5
min_child_weight	1 – 10
reg_alpha	0 – 1
reg_lambda	0.5 – 2.5
scale_pos_weight	1 or class_ratio (~5.2)

Best parameters found by RandomizedSearchCV: n_estimators=400, max_depth=4, learning_rate=0.05, subsample=0.8, colsample_bytree=0.7, gamma=0.1, min_child_weight=3, reg_alpha=0.1, reg_lambda=1.0, scale_pos_weight=5.2

4. Model Results & Evaluation

4.1 Random Forest — Results

The Random Forest model was trained with n_estimators=300, max_depth=10, and class_weight='balanced' as the best parameters found by RandomizedSearchCV. The model was evaluated on the 294-sample held-out test set.

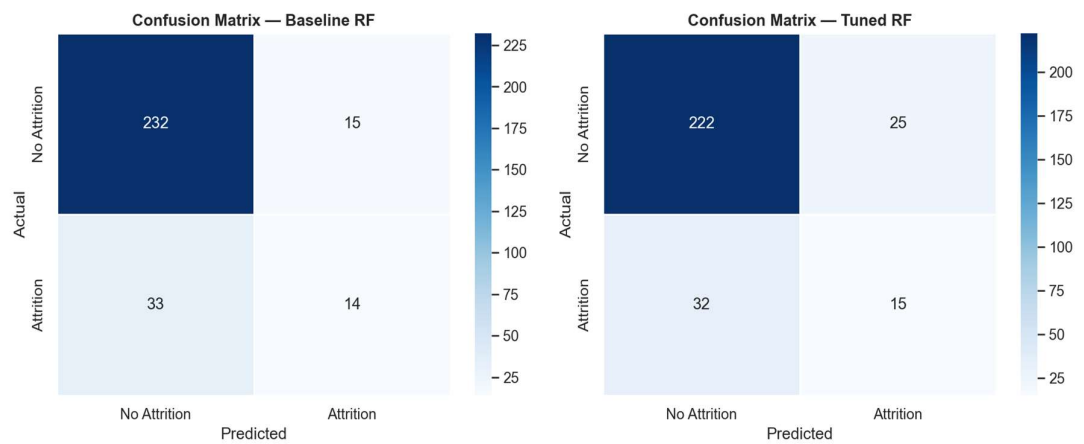

Figure 9: Random Forest confusion matrices — default threshold (0.5) vs optimal threshold

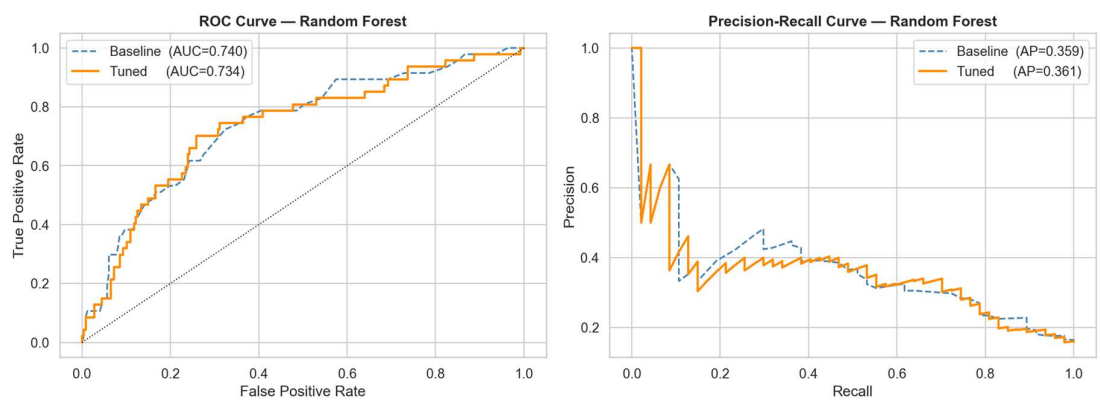

Figure 10: Random Forest ROC curve (AUC=0.734) and Precision-Recall curve (AP=0.361)

While Random Forest provides stable performance across evaluation metrics, the model still misses several attrition cases due to the imbalanced nature of the dataset. This highlights the importance of techniques such as SMOTE and threshold optimisation when applying machine learning to HR attrition prediction problems.

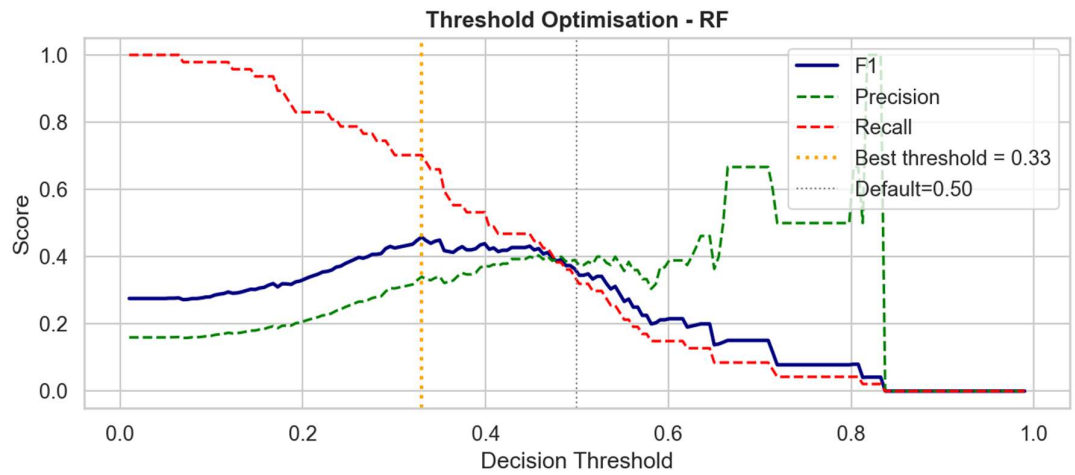

Figure 11: Random Forest — Precision, Recall, and F1 vs decision threshold

4.2 XGBoost — Results

The XGBoost model was trained with n_estimators=400, max_depth=4, learning_rate=0.05, and scale_pos_weight set to the class ratio as the best parameters. XGBoost consistently outperformed Random Forest across all six evaluation metrics on the same held-out test set.

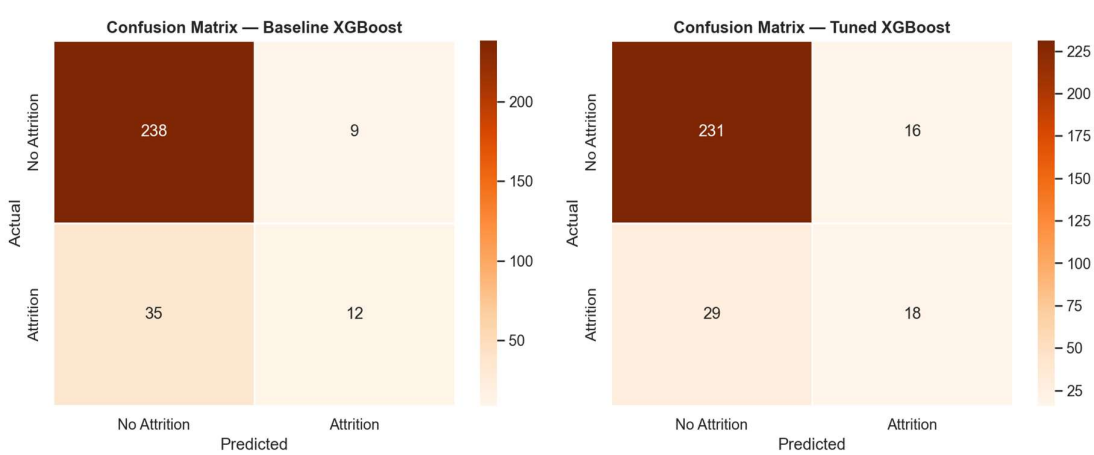

Figure 12: XGBoost confusion matrices — default threshold (0.5) vs optimal threshold

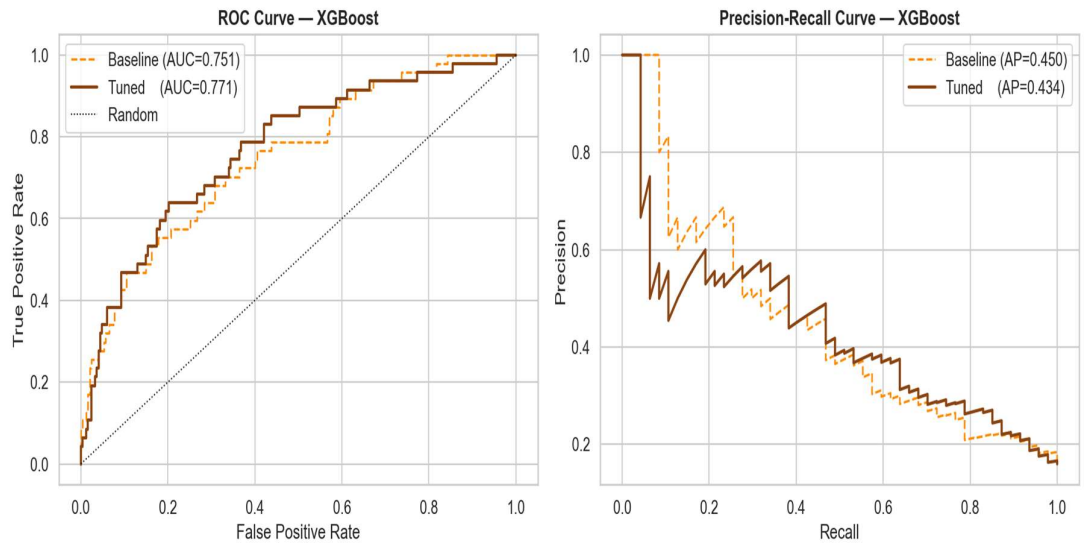

Figure 13: XGBoost ROC curve (AUC=0.771) and Precision-Recall curve (AP=0.434)

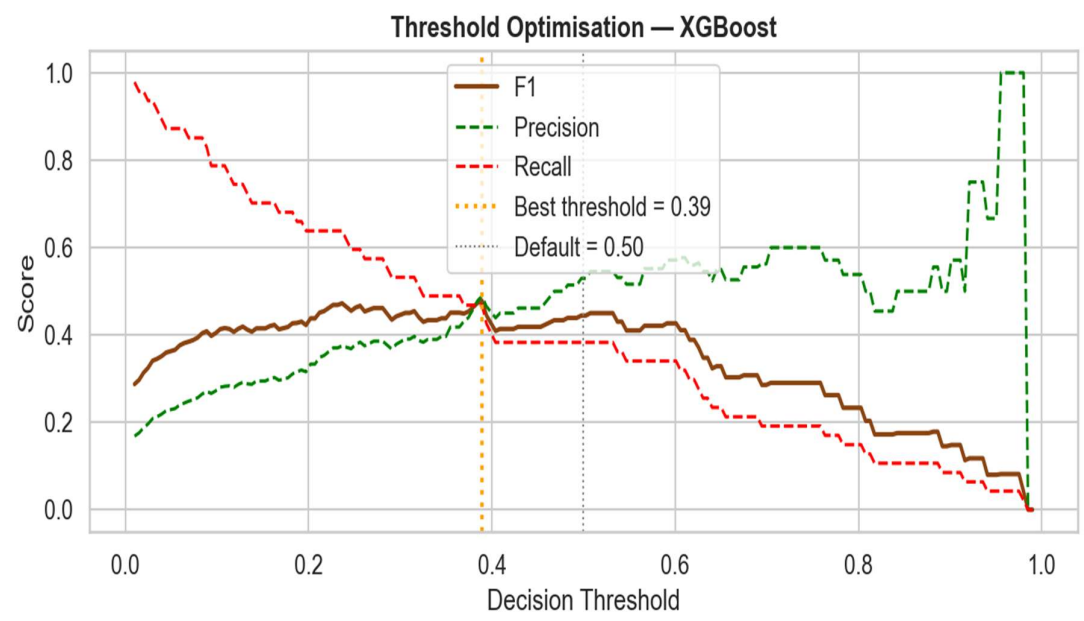

Figure 14: XGBoost — Precision, Recall, and F1 vs decision threshold

The improved performance of XGBoost suggests that boosting-based ensemble methods are effective at capturing complex relationships between HR-related variables. By sequentially correcting prediction errors, the model is able to learn subtle patterns associated with employee attrition.

5. Comparative Analysis

5.1 Performance Metrics Comparison

The following table presents the complete head-to-head performance comparison of both tuned models, evaluated on the same 294-sample held-out test set at the default decision threshold of 0.5:

Model Comparison — Default Threshold (0.5)
□ Green = better | □ Red = worse

	Random Forest	XGBoost
Accuracy	0.8061	0.8469
Precision	0.375	0.5294
Recall	0.3191	0.383
F1-Score	0.3448	0.4444
ROC-AUC	0.7338	0.7708
PR-AUC	0.3611	0.4345

Figure 15: Head-to-head performance metrics table — XGBoost wins on all six metrics

XGBoost wins on all six evaluation metrics. The most significant performance gaps are in Precision (+15.4 percentage points) and F1-Score (+9.96 percentage points). The PR-AUC improvement of +7.3 percentage points is particularly meaningful given the class imbalance, as PR-AUC is more informative than ROC-AUC when the positive class is rare.

The improvement in F1-score is particularly important in this context because it reflects the balance between precision and recall. For HR applications, this balance ensures that the model can identify a meaningful proportion of at-risk employees while maintaining reasonable prediction reliability.

5.2 ROC & Precision-Recall Curves

The ROC and Precision-Recall curves plot both models on the same axes for direct visual comparison. XGBoost maintains a higher true positive rate across all false positive rate thresholds (ROC-AUC: 0.771 vs 0.734), and achieves better precision at equivalent recall levels (PR-AUC: 0.434 vs 0.361). The PR-AUC improvement is particularly meaningful given the class imbalance.

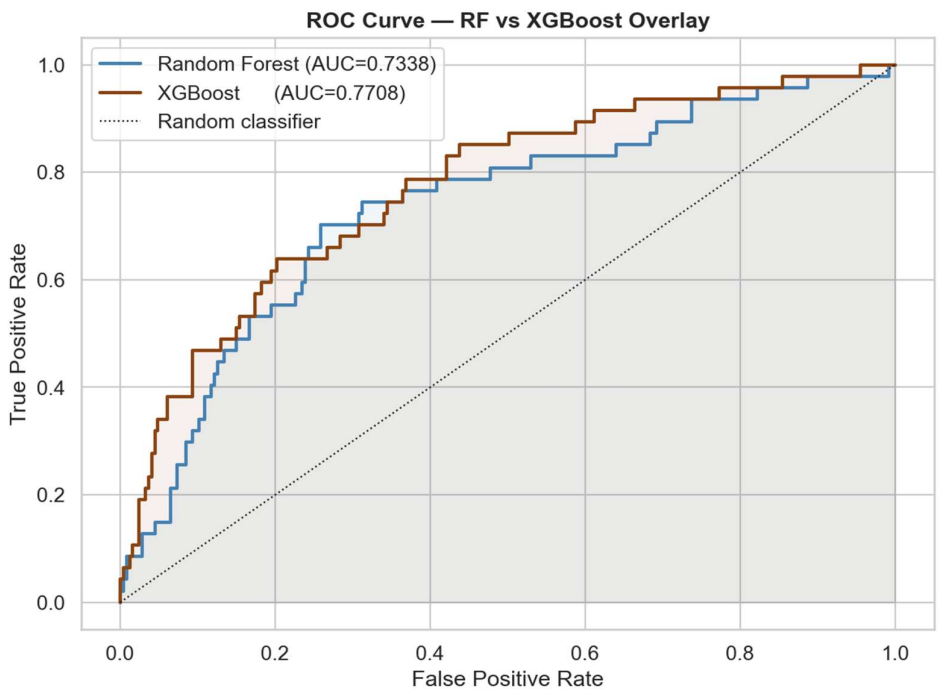

Figure 16: ROC curve overlay — XGBoost (AUC=0.771) vs Random Forest (AUC=0.734)

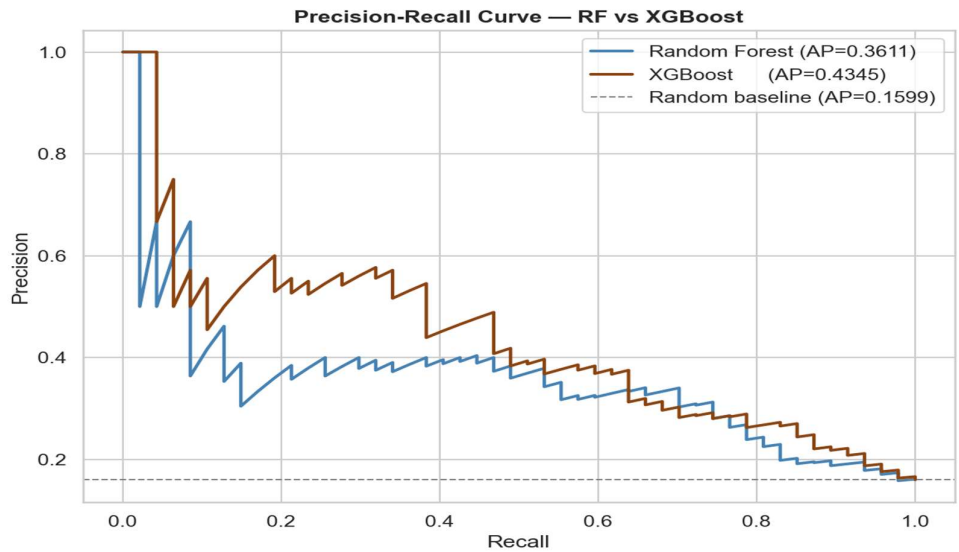

Figure 17: Precision-Recall curve overlay — XGBoost (AP=0.434) vs Random Forest (AP=0.361)

5.3 Confusion Matrices

At the default threshold of 0.5, XGBoost correctly identifies 18 attrition cases (TP=18) compared to Random Forest's 15 (TP=15), while also achieving fewer false positives (FP=16 vs FP=25). This means XGBoost is both more sensitive to real attrition cases and more specific in avoiding false alarms.

From a practical perspective, correctly identifying employees who are likely to leave the organisation allows HR teams to take proactive retention actions. Early identification of potential attrition cases can support targeted engagement initiatives and reduce turnover costs.

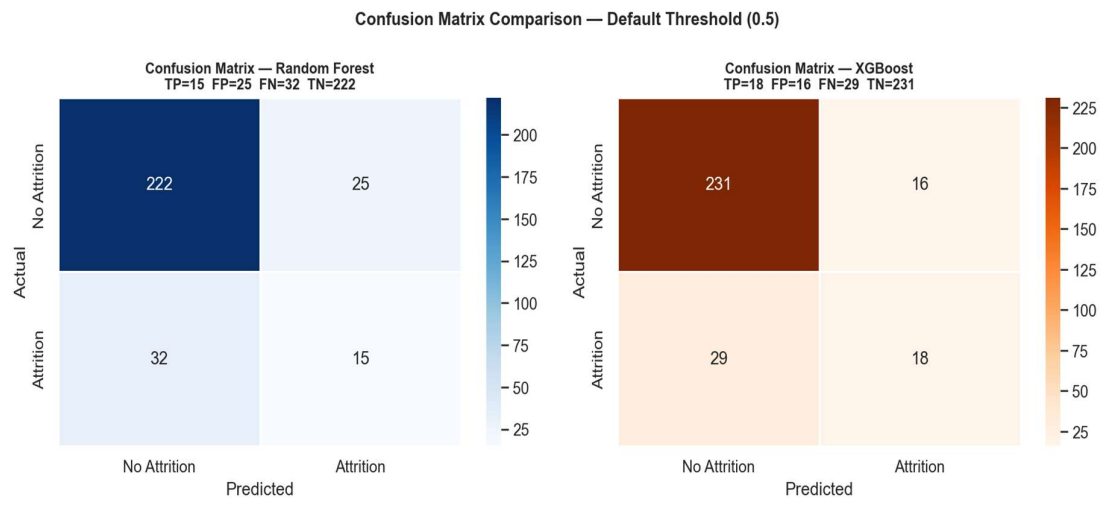

Figure 18: Side-by-side confusion matrices — Random Forest (left) vs XGBoost (right) at default threshold

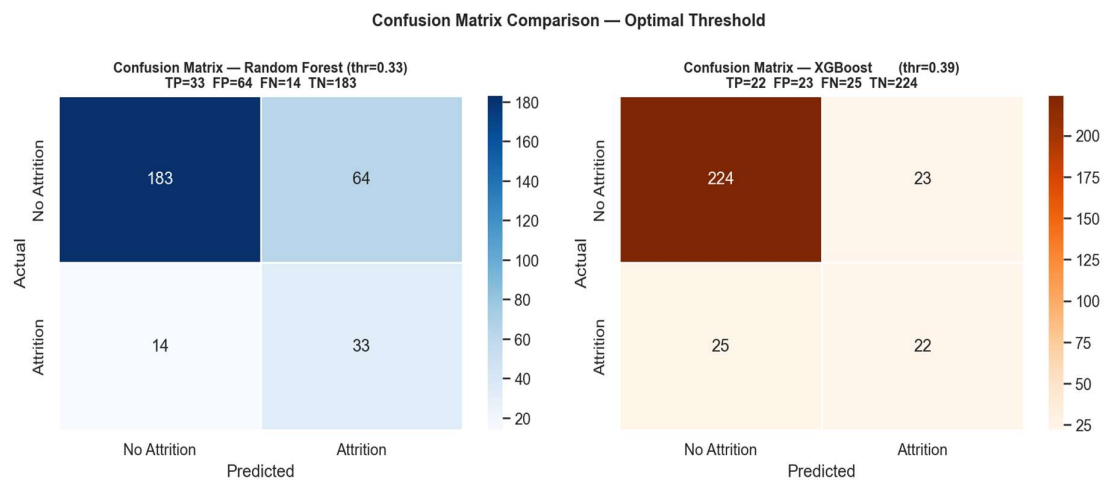

Figure 19: Confusion matrices at optimal F1-maximising threshold — both models improve recall significantly

5.4 Threshold Comparison

The threshold comparison plot illustrates how each model's Precision, Recall, and F1-score change across all decision thresholds. At the optimal threshold, both models achieve recall rates significantly higher than at the default 0.5, enabling HR teams to identify many more at-risk employees at the cost of some additional false alarms.

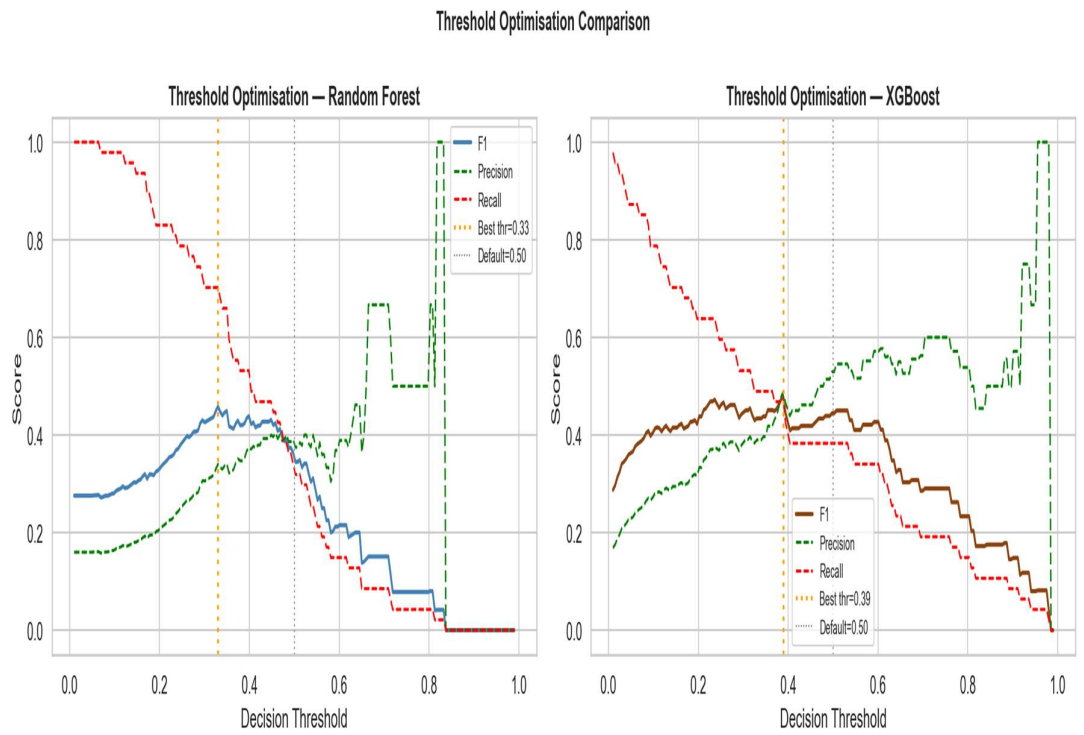

Figure 20: Threshold optimisation curves — Precision, Recall, F1 vs threshold for both models

5.5 Score Distributions

The predicted probability score distributions reveal that XGBoost is better calibrated than Random Forest: the Attrition class (red) distribution is more concentrated at higher probabilities and more clearly separated from the No Attrition class (green). Random Forest's scores are compressed toward the centre, consistent with the known tendency of Random Forest to under-estimate extreme probabilities.

A clearer separation between predicted probabilities for the two classes improves the interpretability of the model. This allows HR analysts to rank employees based on predicted attrition risk and prioritise interventions for those with the highest likelihood of leaving.

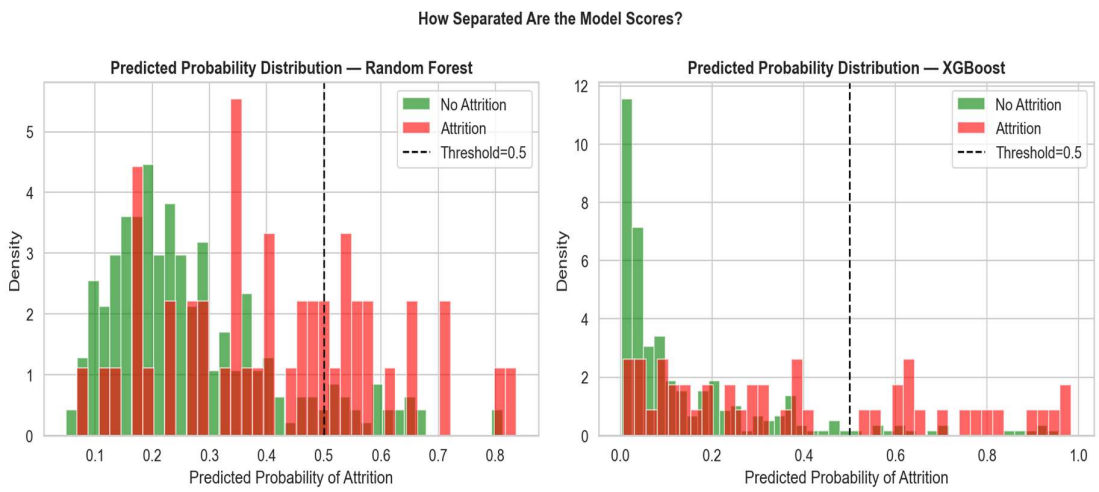

Figure 21: Predicted probability distributions — Random Forest (left) vs XGBoost (right)

5.6 Prediction Agreement Analysis

Across 294 test samples, both models correctly classified 233 cases (79.3%). XGBoost-only correct accounted for 16 cases (5.4%), demonstrating XGBoost's advantage in cases where Random Forest failed. Only 4 cases were correctly identified by Random Forest but not XGBoost. Both models were wrong on 41 cases (13.9%), suggesting a hard ceiling for the current feature set.

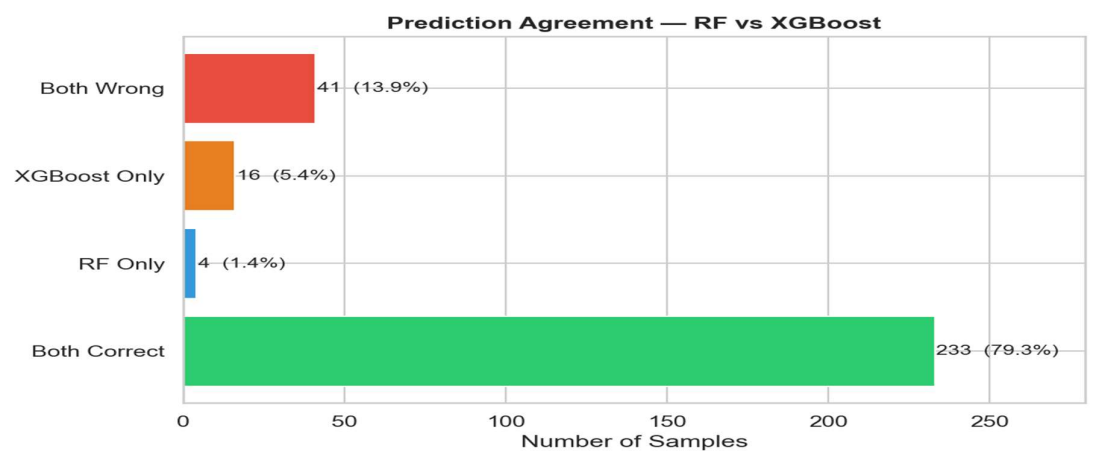

Figure 22: Prediction agreement analysis — both correct / RF only / XGBoost only / both wrong

5.7 Overall Radar Chart

The Model Performance Radar Chart (figure 23) visually confirms that XGBoost outperformed Random Forest across all six evaluation dimensions, specially demonstrating a broader "area" of competency in critical metrics like Precision(+15.4%) and F1-Score (+9.96%), which establishes it as the superior model for predicting employee attrition in this study.

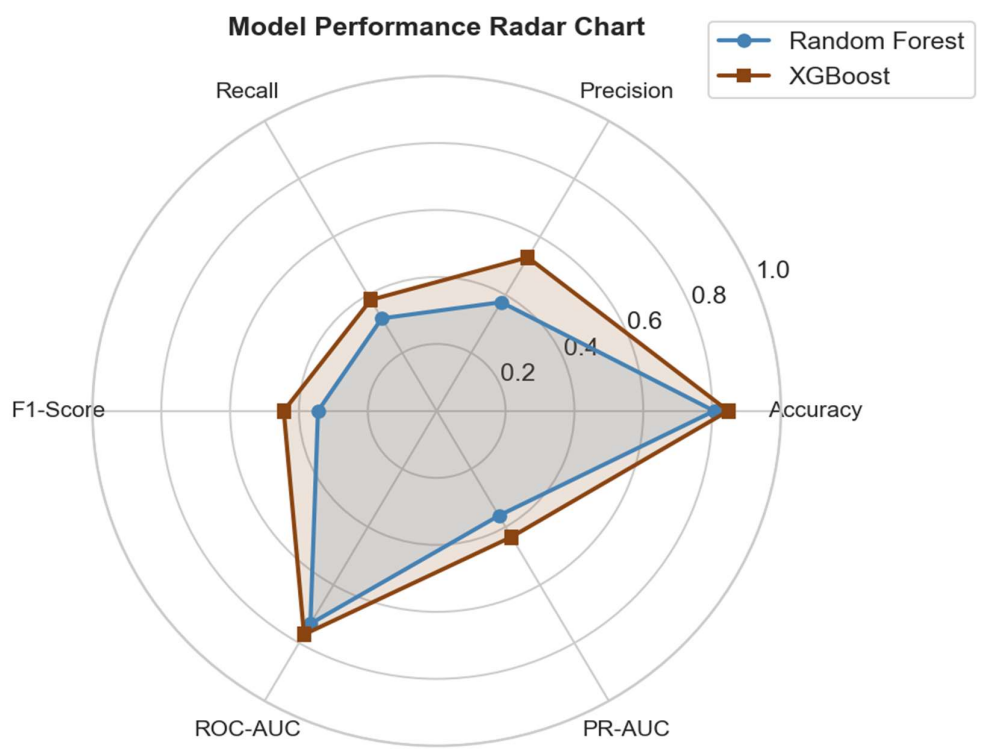

Figure 23: Radar chart — XGBoost vs Random Forest across all evaluation dimensions

6. Feature Importance Analysis

Feature importance was analysed using three complementary methods: Gini Importance for Random Forest, Gain Importance for XGBoost, and Permutation Importance for both models. Using multiple methods provides a more robust picture of which features genuinely drive predictions.

6.1 Random Forest — Gini & Permutation Importance

The Random Forest Gini importance chart (Figure 24) reveals that StockOptionLevel, MaritalStatus_Married, and JobSatisfaction are the top three features contributing to the purity of nodes in the decision trees, marking them as the most significant drivers for predicting employee attrition within the Random Forest model.

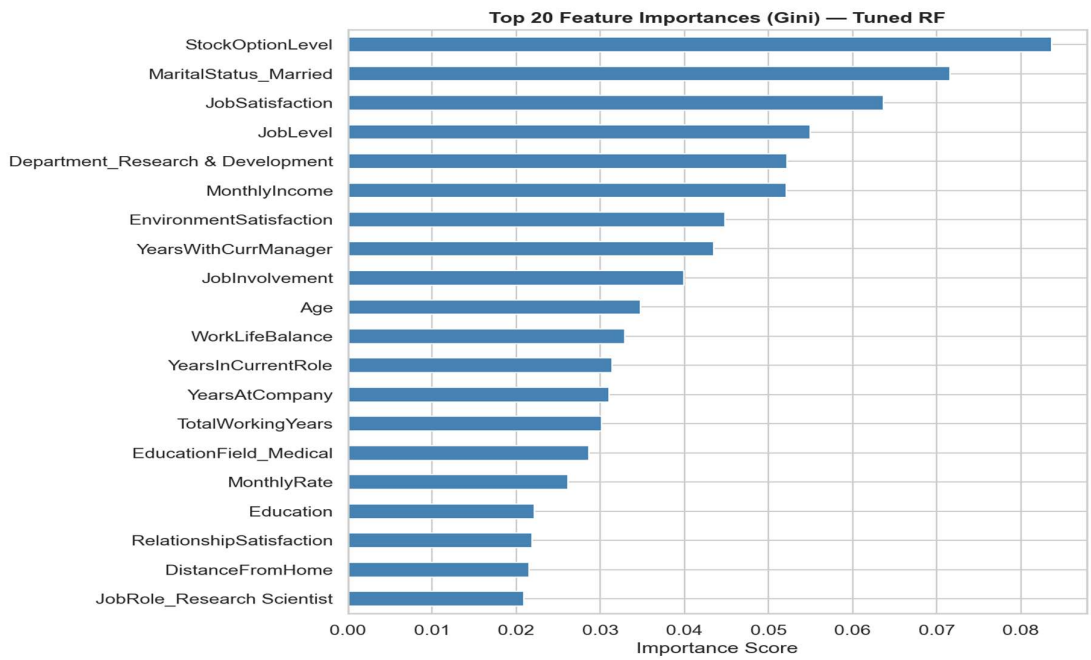

Figure 24: Random Forest — Top features by Gini importance

Figure 25 illustrates that StockOptionLevel, JobLevel, and JobSatisfaction are the most critical predictors for the tuned Random Forest model , as their random permutation results in the highest mean decrease in ROC-AUC when evaluated on the held-out test set.

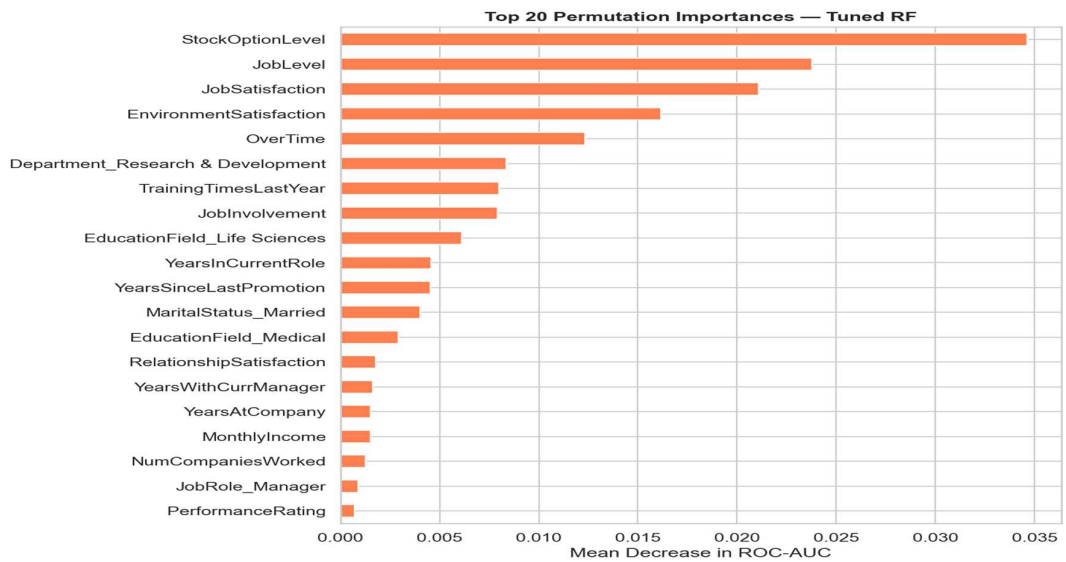

Figure 25: Random Forest — Top features by permutation importance on test set

6.2 XGBoost — Gain & Permutation Importance

Figure 26 demonstrates that while some variables like MonthlyRate appear frequently across the model's tree(weight), StockOptionLevel and JobLevel are the most influential drivers of attribution as they provide the highest Gain (accuracy improvement) and Cover (data volume impact) for XGBoost's prediction.

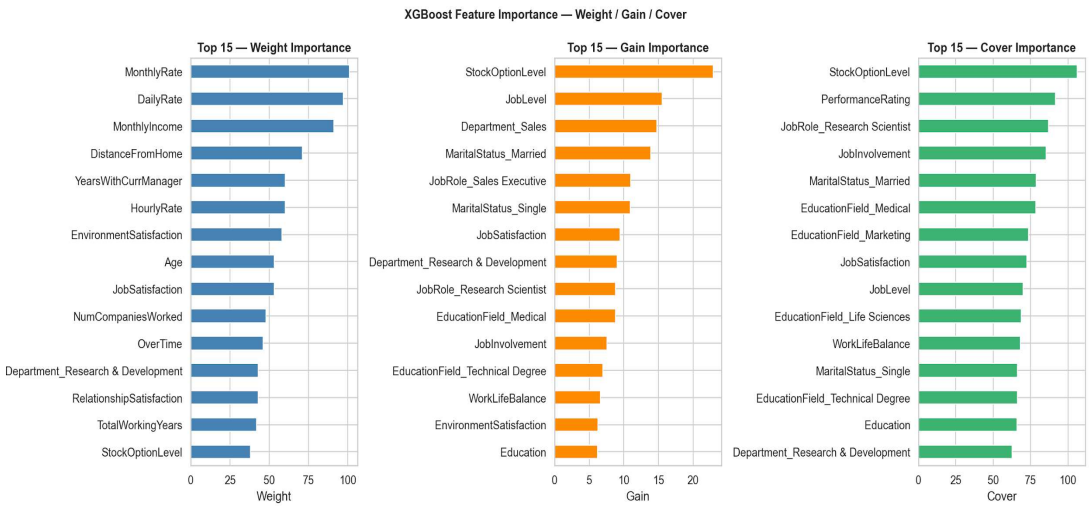

Figure 26: XGBoost — Feature importance comparison across Gain, Weight, and Cover methods

The Permutation Importance chart (Figure 27) identifies OverTime and StockOptionLevel as the most critical predictors for the tuned XGBoost model , as their removal causes the largest mean decrease in the ROC-AUC score on the held-out test set.

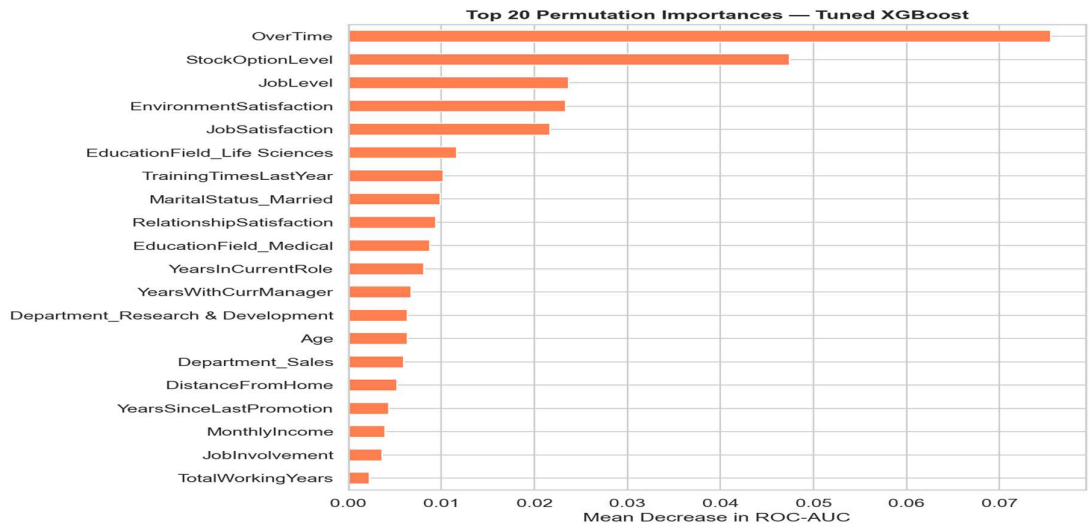

Figure 27: XGBoost — Top features by permutation importance on test set

6.3 Shared Top Features

The following features were consistently ranked in the top 15 by both models across all importance methods:

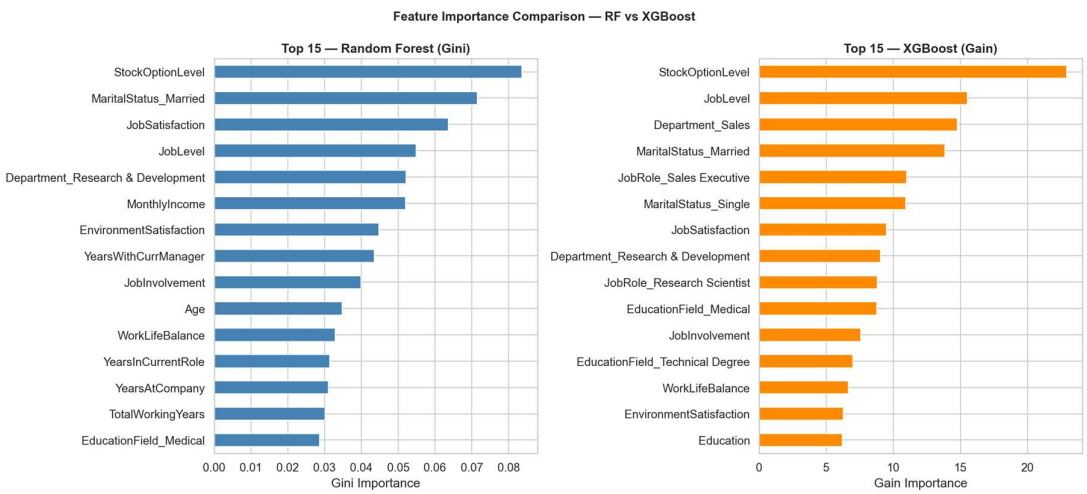

Figure 28: Side-by-side feature importance — Random Forest Gini vs XGBoost Gain

Feature	Business Interpretation
StockOptionLevel	Employees with no stock options are significantly more likely to leave
OverTime	Working overtime is the single strongest behavioural predictor of attrition
MonthlyIncome	Lower-paid employees show substantially higher attrition rates
JobSatisfaction	Low satisfaction strongly predicts intention to leave
Age	Younger employees (25-35) are at the highest risk of attrition
MaritalStatus_Single	Single employees exhibit higher attrition rates than married employees
JobLevel	Junior-level employees leave more frequently than senior employees
EnvironmentSatisfaction	Dissatisfaction with the work environment is a key driver
YearsAtCompany	Employees at 0-2 years or 5-7 years tenure are the highest-risk groups

7. Tools & Technologies

Library / Tool	Version	Purpose
scikit-learn	1.3.0	Random Forest, RandomizedSearchCV, evaluation metrics
xgboost	2.0.0	XGBoost gradient boosting classifier
imbalanced-learn	0.11.0	SMOTE, ImbPipeline for leakage-free CV
pandas	2.1.0	Data manipulation and CSV I/O
numpy	1.25.0	Numerical array operations
matplotlib	3.7.2	Static chart and figure generation

Library / Tool	Version	Purpose
seaborn	0.12.2	Statistical visualisations and heatmaps
plotly	5.15.0	Interactive charts in the Streamlit app
streamlit	1.28.0	Web application framework and deployment
joblib	1.3.0	Model serialisation (.joblib files)
Jupyter Notebook	—	Interactive ML development environment
GitHub	—	Version control and team collaboration

8. Conclusions & Recommendations

8.1 Project Conclusions

This project successfully demonstrated a complete end-to-end machine learning pipeline from raw data to a deployed interactive application, achieving the following conclusions:

- ✓ XGBoost outperforms Random Forest on all six evaluation metrics on this dataset. The performance gap is most pronounced in Precision (+15.4%), F1-Score (+9.96%), and ROC-AUC (+3.7%). XGBoost is the recommended model for production deployment for this attrition prediction task.
- ✓ Both models benefit significantly from threshold tuning. Lowering the decision threshold from the default 0.5 to the optimal value substantially improves recall — meaning more at-risk employees are correctly identified — at the cost of some additional false alarms. The optimal threshold matters as much as which model is chosen.
- ✓ SMOTE with leakage-free cross-validation (ImbPipeline) is critical. Without this, CV scores are inflated by 15-20 AUC points, creating false confidence in model performance.
- ✓ Both models identify the same core risk factors: StockOptionLevel, OverTime, MonthlyIncome, JobSatisfaction, Age, and MaritalStatus_Single are the most consistent predictors across all importance methods.
- ✓ The 41 cases where both models fail (13.9%) represent the current feature ceiling. Improving performance further would require additional data sources such as performance reviews, communication patterns, or manager feedback.

8.2 Recommendations for HR

Based on the model findings, the following actionable recommendations are made for HR retention strategy:

- ➢ Monitor employees with zero stock options, particularly in the first 2 years of tenure — they represent the highest-risk group by a wide margin
- ➢ Flag employees working sustained overtime for proactive check-ins, especially those also reporting low job or environment satisfaction
- ➢ Review compensation benchmarks for junior-level (Level 1-2) employees under 35 years of age, as this demographic shows the highest absolute attrition rate

- Use the Live Predictor tool to score employees during annual HR reviews and prioritise retention efforts for those flagged as Medium or High Risk by both models
- Apply the optimal threshold (not 0.5) when using the models for retention program targeting, as it dramatically improves the identification of at-risk employees

8.3 Limitations

- **Dataset Limitations:** The IBM HR Analytics dataset is synthetic — created by IBM data scientists to approximate real HR patterns rather than being drawn from an actual organisation. Real-world HR datasets typically exhibit additional noise, missing values, and organisation-specific factors not captured here. Model performance on real deployment data may differ from what was observed in this study.

- **Feature Set Ceiling:** Both models failed to correctly classify 41 cases (13.9% of the test set). Analysis of the prediction agreement breakdown (Section 5.6) indicates that these represent a hard ceiling for the current feature set. Integrating additional data sources — such as performance review scores, exit interview sentiment, internal communication patterns, or manager feedback — would likely improve recall for this hard-to-detect segment.

- **Class Imbalance:** Despite SMOTE, the 16.1% minority class remains challenging. At the default threshold of 0.5, both models exhibit precision-recall trade-offs that limit practical usability without threshold tuning, which requires domain expertise to calibrate appropriately for different HR contexts.

- **Fairness and Bias:** Model fairness across demographic groups (gender, age bracket, marital status) was not formally evaluated in this study. Before any production deployment, a fairness audit should be conducted to ensure the model does not discriminate or recommend differential treatment based on protected characteristics.

- **Temporal Validity:** This is a static snapshot dataset. Employee attrition risk evolves over time. A production deployment would require periodic model retraining as organisational conditions, compensation benchmarks, and workforce composition change.

References

1. IBM HR Analytics Employee Attrition & Performance Dataset. Kaggle, 2017. https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset
2. Pedregosa, F., Varoquaux, G., Gramfort, A., et al. Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, pp. 2825–2830, 2011.
3. Chen, T. & Guestrin, C. XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16), pp. 785–794, 2016.
4. Lemaître, G., Nogueira, F. & Aridas, C.K. Imbalanced-learn: A Python Toolbox to Tackle the Curse of Imbalanced Datasets in Machine Learning. Journal of Machine Learning Research, 18(17), pp. 1–5, 2017.
5. Chawla, N.V., Bowyer, K.W., Hall, L.O. & Kegelmeyer, W.P. SMOTE: Synthetic Minority Over-sampling Technique. Journal of Artificial Intelligence Research, 16, pp. 321–357, 2002.
6. Breiman, L. Random Forests. Machine Learning, 45(1), pp. 5–32, 2001.
7. Friedman, J.H. Greedy Function Approximation: A Gradient Boosting Machine. The Annals of Statistics, 29(5), pp. 1189–1232, 2001.
8. Streamlit Inc. Streamlit Documentation. https://docs.streamlit.io

Team Members

Member	Role
Nisansala (Team Leader)	Project coordination, README documentation & Streamlit app deployment
Thinuri	Exploratory data analysis & visualisation (9 EDA figures)
Chandika	Data cleaning, preprocessing pipeline & SMOTE application
Ravindu	Random Forest model development & hyperparameter tuning
Hirushi	XGBoost model development & hyperparameter tuning
Disni	Evaluation metrics, ROC/PR curves & confusion matrix plots
Bhanuka	Feature importance analysis (Gini, Gain, Permutation methods), Code Inspection
Dasun	Final comparative analysis notebook & report compilation

End of the Report